

Agentic Model Predictive Control: Operating in Intelligence Space via Cognitive Supervisory Layers and Parallel Path Integral Rollouts — A Simulation Case Study

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ABSTRACT

Model Predictive Control (MPC) has served as the industrial benchmark for constrained multivariable control across process operations for four decades. Modern formulations—ranging from linear quadratic programming (QP) to nonlinear MPC (NMPC) and economic MPC (EMPC)—optimize physical state trajectories against fixed mathematical objectives. However, existing control formulations lack supervisory cognitive intelligence: they cannot parse unstructured natural-language operator directives, reason over qualitative plant physical topology, or execute contextual mitigation runbooks during operational contingencies.

In this paper, we propose **Agentic Model Predictive Control (Agentic MPC)**, an architecture that places an **Intelligence Space** supervisory layer above high-throughput real-time control solvers. The proposed architecture partitions supervisory intelligence into **The Two Halves**: (1) a neural reasoner (*The Knower*) grounded in four structured memory tiers (Playbook, Rulebook, Yearbook, and Whiteboard), and (2) a deterministic Executive Function Harness (*The Doer*) enforcing Ring 0 operating invariants, automated contract unit tests, and preemptive trip bounds. In our architecture, high-level cognitive directives are mapped into parameterized Model Predictive Path Integral (MPPI) cost manifolds executed across parallel trajectory rollouts via GPU/WebGPU compute shaders, supported by an online lagged parameter filter.

We evaluate the physical dynamics on an exothermic Continuous Stirred-Tank Reactor (CSTR) undergoing non-linear kinetic surges across a fully reproducible numerical simulation case study spanning 20 randomized scenarios ($C_{A0} \in [+10\%, +30\%]$, $T_0 \in [+5 \text{ K}, +12 \text{ K}]$, and $UA \in [-10\%, -35\%]$) evaluated over a standardized 60.0 s benchmark window with advance-warning advisory lead time ($t_{\text{lead}} = 3.5 \text{ s}$). In all preview evaluations, controllers receive an exact disturbance forecast (both onset timing and surge magnitude); reported preview results thus represent a perfect-forecast upper bound, while forecast timing jitter and magnitude errors remain untested. Numerical simulation establishes an **empirical feedback floor of 6/20 trips across feedback controllers on this scenario set**: when tuned equally ($q_T = 10.0$), Linear MPC trips in 6/20 scenarios ($2.26 \pm 1.14 \text{ K}$ non-trip RMSE, $9.49 \pm 5.39 \text{ K}$ overall RMSE), matching NMPC (6/20 trips, $2.48 \pm 1.52 \text{ K}$ non-trip, $9.62 \pm 5.37 \text{ K}$ overall) and MPPI with a lagged-UA filter (6/20 trips, $2.61 \pm 1.34 \text{ K}$ non-trip, $9.82 \pm 5.39 \text{ K}$ overall), all tripping on the identical six scenarios (Scenarios 4, 5, 8, 16, 18, 20). When weakly penalized ($q_T = 1.0$), Linear MPC trips in 12/20 scenarios ($16.26 \pm 5.52 \text{ K}$ overall RMSE). Notably, all reactive trips and the single preview trip occur 4–31 s after the surge ends during post-surge recovery, rather than during surge onset.

When advance preview lookahead ($t_{\text{lead}} = 3.5 \text{ s}$) is supplied, trips drop to 1/20 across all preview controllers, with failure occurring exclusively on Scenario 18 (an extreme corner realization requiring 6.0 s of advance lead in open loop, the sole scenario requiring $> 5.0 \text{ s}$). Crucially, our ablation isolates that the supervisory schedule's quantitative advantage ($0.94 \pm 0.41 \text{ K}$ non-trip RMSE, $2.15 \pm 2.55 \text{ K}$ overall RMSE vs. $1.75 \pm 1.00 \text{ K}$ non-trip, $3.10 \pm 2.98 \text{ K}$ overall for standard preview MPPI, representing a 0.81 K gain across 19 shared survivors, paired t -test $p = 0.013$, but $p = 0.063$ after Holm-Bonferroni correction, rendering this gain suggestive rather than confirmed) stems from coordinating an anticipatory pre-cooling ramp that is consistent with avoiding **reaction quenching**: naive rule-based step switching quenches the reactor ($T_{\text{min}} = 339\text{--}344 \text{ K}$), accumulates unreacted feed ($C_A = 0.555\text{--}0.576 \text{ mol/L}$ at surge conclusion, climbing post-surge to $0.578\text{--}0.597 \text{ mol/L}$), and triggers delayed thermal blowout (8/20 trips, $1.45 \pm 0.68 \text{ K}$ non-trip RMSE). We present the LLM supervisory layer as the proposed architecture for qualitative interaction and formulate the physical control and safety contract harness, establishing a simulation case study for anticipatory process control.

1. Introduction & Related Work

Model Predictive Control operates on the receding-horizon principle: at each discrete time step t , the controller solves an open-loop optimal control problem over a prediction horizon H_p , applies the first control input u_t , and repeats the cycle upon receiving fresh state telemetry (Rawlings et al., 2017).

While classical linear MPC remains computationally tractable on millisecond timescales via convex quadratic programming (QP), linear approximations degrade on highly non-linear chemical plants governed by exponential kinetics, such as Continuous Stirred-Tank Reactors (CSTR) undergoing Arrhenius heat generation (Seborg et al., 2016). When state perturbations depart from the nominal linearization point, purely reactive controllers exhibit severe tracking error, control chattering, and valve saturation.

1.1 Related Work: Advanced & Learning-Based MPC

- **Robust and Constrained MPC:** Linear Matrix Inequality (LMI) and tube-based robust MPC formulations synthesize invariant sets to maintain constraint satisfaction under bounded disturbances (Kothare et al., 1996; Mayne et al., 2005).
- **Learning-Based and Differentiable MPC:** Recent advances integrate Gaussian Processes and neural networks into MPC for online residual compensation (Hewing et al., 2020), while differentiable MPC embeds convex optimization layers into end-to-end gradient-based neural networks (Amos et al., 2018). Physics-informed parameter tracking formulations have motivated domain-constrained residual filters that track conservation laws during online estimation; in this study, parameter compensation is evaluated via a direct lagged parameter tracking filter.
- **Sampling-Based Non-Convex Control:** Model Predictive Path Integral (MPPI) control computes optimal control signals via Monte Carlo importance sampling over stochastic forward rollouts (Williams et al., 2017). Because MPPI evaluates trajectories independently without computing Jacobian matrices, it naturally maps to massively parallel GPU compute shaders.

1.2 The Role of the Supervisory Layer: Lookahead vs. Reasoning

A critical insight in receding-horizon control is that numerical solvers optimize over a fixed finite horizon H_p (e.g., 4.0 s). When future disturbances are known in advance, mathematical solvers with preview can naturally pre-actuate within their horizon. In our benchmark simulations, preview controllers receive the exact disturbance profile (timing and magnitude), representing an idealized perfect-forecast upper bound.

However, in actual plant operations, forecasts arrive as human shift handover notes, laboratory feed analysis reports, or upstream DCS alarms. Classical MPC cannot parse these qualitative inputs.

Agentic MPC addresses this supervisory gap. Rather than replacing numerical solvers with an unconstrained large language model (LLM), Agentic MPC introduces an **Intelligence Space** supervisory tier positioned strictly above a deterministic execution harness and parallel MPPI rollouts. The neural reasoner is designed to interpret operator directives and contextual operational runbooks, propose quantitative cost manifolds, and verify them through deterministic contract tests before any parameter update reaches physical actuators.

2. The Agentic MPC Architecture

2.1 The Two Halves: The Knower and The Doer

- **The Knower (Neural Reasoner):** Operates in *Intelligence Space*, translating operator directives into candidate cost manifold parameters $\theta_{\mathcal{M}} = \{q_{\text{temp}}, q_{\text{barrier}}, u_{\text{target}}, \lambda\}$. Inference runs on dedicated cloud LPUs or server GPUs (e.g., Qwen-2.5-32B), with lightweight client-side WebGPU shaders (e.g., Qwen-2.5-0.5B via @mlc-ai/web-llm) available for offline edge operation.
- **The Doer (Executive Function Harness):** A deterministic verification kernel enforcing Ring 0 invariants (actuator saturation and slew rate limits), running automated contract unit tests, and operating preemptively below plant emergency shutdown thresholds.

2.2 The Four Structured Memory Subsystems

Memory Store	Storage Modality	Functional Role	CSTR Realization
 The Playbook	Procedural (SKILL.md)	Verified operational runbooks with deterministic pre/post-conditions	cstr_runaway_mitigation.md verified via automated contract tests
 The Rulebook	Semantic (Vector Store)	Safe operating envelopes and regulatory engineering constraints	Safe operating limit envelope: $T_{\text{trip}} = 385.0$ K, nominal $T_{\text{ref}} = 350.0$ K
 The Yearbook	Relational (Graph DB)	Plant equipment topological connectivity and flow dependencies	Piping & Instrumentation: V-101 → J-101 → CV-201 → M-101 → CH-3
 The Whiteboard	Shared State (IPC)	Real-time working scratchpad with distributed mutex locking	Mutex locks (MUTEX_PRECOOL) preventing conflicting concurrent directives

(Note: While these structured memory tiers define the conceptual supervisory architecture of Agentic MPC, they are not directly exercised in the numerical CSTR benchmark experiments reported below, which evaluate the deterministic control and anticipatory scheduling layer.)

3. Mathematical Formulation

3.1 Classical Linear MPC Baseline (Jacobian QP)

The linear discrete-time optimal control problem is solved as a condensed Quadratic Program (QP) around the textbook steady state ($C_{A,\text{ref}} = 0.50$ mol/L, $T_{\text{ref}} = 350.0$ K, $T_{c,\text{base}} = 300.0$ K):

$$\min_U \frac{1}{2} U^T H_{\text{qp}} U + g^T U \quad \text{subject to: } u_{\min} \leq u_k \leq u_{\max}, \quad |\Delta u_k| \leq \Delta u_{\max}$$

The continuous Jacobian matrix A has eigenvalues $[-0.0076, +0.0472]$ s⁻¹, exhibiting an open-loop thermal runaway pole ($\tau_{\text{growth}} \approx 21.2$ s). State penalty is $Q = \text{diag}(10.0, q_T)$, $R = 0.02$, input bounds $[280, 360]$ K, and slew limit 12 K/s. We evaluate Linear MPC at both $q_T = 1.0$ and $q_T = 10.0$.

3.2 Model Predictive Path Integral (MPPI) Formulation

MPPI evaluates $K = 1,024$ stochastic control rollouts sampled from $\mathcal{N}(0, \Sigma)$ with covariance $\Sigma = 6.0^2 I$ and temperature $\lambda = 10.0$. Discretized at rollout step $\Delta t_{\text{ctrl}} = 0.2$ s:

$$x_{k+1} = f(x_k, v_k)$$

$$S(U^{(m)}) = \sum_{k=0}^{H_p-1} (q_{\text{temp}}(T_k - T_{\text{ref}})^2 + q_{\text{barrier}} \max(0, T_k - T_{\text{barrier}})^2)$$

(Note: In our numerical implementation, the standard theoretical control-cost term $\frac{\lambda}{2} \sum_{k=0}^{H_p-1} \epsilon_k^T \Sigma^{-1} \epsilon_k$ is omitted, evaluating state tracking and barrier penalties directly over the candidate rollouts.)

The optimal control update is computed via softmax importance weighting:

$$u_t^* = u_t + \sum_{m=1}^K w(U^{(m)}) \epsilon_t^{(m)}, \quad w(U^{(m)}) = \frac{\exp(-\frac{1}{\lambda}(S(U^{(m)}) - \min_j S(U^{(j)})))}{\sum_{k=1}^K \exp(-\frac{1}{\lambda}(S(U^{(k)}) - \min_j S(U^{(j)})))}$$

3.3 Lagged-UA Tracking Filter

To evaluate parameter identification during fouling, a first-order lagged filter tracks the true parameter $UA(t)$ after surge onset:

$$UA_{\text{est}} \leftarrow UA_{\text{est}} + 0.02(UA_{\text{true}} - UA_{\text{est}})$$

evaluated at each integration step $\Delta t_{\text{sim}} = 0.02$ s ($\tau = 1.0$ s). Uses the true parameter directly. With forecast lookahead, the forecast vector supplies UA directly, so the filter primarily impacts the no-forecast baseline (12.13 K $\rightarrow$ 9.82 K overall RMSE).

4. Benchmark Case Study: Exothermic CSTR

4.1 Hold-280 K Lead-Time Sweep & Controllability Limits

An empirical **hold-280 K lead-time sweep** over the benchmark kinetic surge evaluates required early advisory warning when stepping jacket coolant flatly to $u = 280$ K. When coolant is released open-loop back to 300 K at surge conclusion ($t = 25$ s), reactant accumulated during deep pre-cooling causes delayed re-ignition at $t = 57\text{--}83$ s unless ≥ 5.0 s lead is provided:

Pre-Cooling Lead Time t_{lead}	Peak Temp T_{max} (60 s)	Peak Temp T_{max} (200 s)	SIS Trip Status (≥ 385.0 K)	Measured Peak Time t_{peak} (Post-Surge Delay)
0.0 s (At Onset)	447.0 K	447.0 K	TRIP (Breached)	$t = 37.2$ s (12.2 s post-surge)
1.0 s Lead Time	445.6 K	445.6 K	TRIP (Breached)	$t = 40.1$ s (15.1 s post-surge)
2.0 s Lead Time	444.0 K	444.0 K	TRIP (Breached)	$t = 44.6$ s (19.6 s post-surge)
2.5 s Lead Time	443.1 K	443.1 K	TRIP (Breached)	$t = 47.7$ s (22.7 s post-surge)
3.0 s Lead Time	442.2 K	442.2 K	TRIP (Breached)	$t = 51.7$ s (26.7 s post-surge)
3.25 s Lead Time	441.7 K	441.7 K	TRIP (Breached)	$t = 54.2$ s (29.2 s post-surge)
3.5 s Lead Time	441.2 K	441.2 K	TRIP (Breached)	$t = 57.3$ s (32.3 s post-surge)
4.0 s Lead Time	375.7 K	440.0 K	TRIP (at 200 s)	$t = 65.8$ s (40.8 s post-surge, delayed ignition)
4.5 s Lead Time	355.0 K	438.5 K	TRIP (at 200 s)	$t = 82.1$ s (57.1 s post-surge, delayed ignition)
5.0 s Lead Time	350.0 K	350.0 K	SAFE (Zero Trip)	Clamped at nominal setpoint ($T < 350.5$ K)

Mechanisms & Per-Scenario Requirements: Flat pre-cooling to 280 K is an elementary heuristic policy. In open-loop release back to 300 K without feedback regulation, unreacted feed that accumulated during deep 280 K cooling re-ignites after surge termination, pushing peak temperatures above 438 K at $t = 65\text{--}83$ s. Only ≥ 5.0 s lead achieves complete thermal exhaustion in open loop for the nominal disturbance scenario.

Evaluation Horizon & Timing Resolution: Open-loop lead times are simulated at 50 Hz ($\Delta t_{\text{sim}} = 0.02$ s), exactly matching the physical plant dynamics and `benchmark_cstr_unified.py --oracle`. Release back to 300 K occurs at $t = 25.0$ s. In contrast, closed-loop benchmark runs in Section 6 evaluate over a standardized 60.0 s horizon.

Per-Scenario Open-Loop Lead Requirements (200 s Horizon): Scenario 18 requires **6.0 s** of lead time (the sole scenario requiring > 5.0 s); Scenarios 8 and 20 require **4.5 s**; Scenarios 4 and 5 require **4.0 s**; Scenarios 7, 15, and 9 require **2.5 s, 2.5 s, and 1.5 s**; and Scenarios 11 and 13 require **0.0 s**.

Rule Supervisor Scenario 18 Survival: Notably, the naive rule supervisor (stepping cooling to 280 K throughout alert and surge, followed by MPPI feedback) safely survives Scenario 18 ($T_{\text{min}} = 348.8$ K, $T_{\text{max}} < 385$ K), whereas the Full anticipatory system and standard preview MPPI trip on Scenario 18 at 3.5 s lead. This highlights that extreme continuous cooling can survive severe kinetic excursions, but at the expense of quenching trips across moderate scenarios (e.g. Scenarios 2, 3, 9).

5. Verification & Safety Architecture

Agentic MPC enforces defense-in-depth: Ring 0 invariant checking ($q_T \geq 0.1$, $q_{\text{barrier}} \geq 1000$, $u \in [280, 360]$ K, slew ≤ 12 K/s), automated contract unit tests on the active fouled plant model (<30 ms), a software preemptive breaker at 382 K, and an independent hardware SIS scram at 385 K (failsafe cooling valve full open, feed shut off).

(Note: The 382 K preemptive software breaker, automated contract unit test suite (< 30 ms), and Ring 0 invariant gating represent the proposed supervisory safety architecture; these software supervisor safeguards are not exercised in the numerical benchmark runs in Section 6, where safety trips are triggered exclusively by the plant SIS limit at 385 K.)

Metric Definition: Overall RMSE includes the post-trip SCRAM transient and depends heavily on the emergency shutdown model. Non-trip RMSE and trip counts serve as the primary metrics.

6. Empirical Results & Ablation Analysis

6.1 Multi-Tier Benchmark Across 20 Randomized Scenarios

Table 2 presents reproducible numerical simulation results across 20 randomized scenarios computed directly from discrete QP, L-BFGS-B NMPC, and NumPy MPPI path integral solvers:

Configuration	Controller Type	Non-Trip RMSE (K)	Overall RMSE (K)	Peak T (K)	Slew (K/s)	SIS Trips	Settled
Baseline 1a	Linear MPC ($q_T = 1.0$, No Forecast)	2.56 ± 1.54	16.26 ± 5.52	389.3 ± 14.2	3.29 ± 0.95	12 / 20 (60%)	8 / 20 (9.8 s)
Baseline 1b	Linear MPC ($q_T = 10.0$, No Forecast)	2.26 ± 1.14	9.49 ± 5.39	372.8 ± 13.8	5.50 ± 1.35	6 / 20 (30%)	13 / 20 (8.5 s)
Baseline 2a	NMPC (L-BFGS-B, No Forecast)	2.48 ± 1.52	9.62 ± 5.37	372.9 ± 13.7	5.04 ± 1.29	6 / 20 (30%)	13 / 20 (6.9 s)
Ablation 1	MPPI (Static θ^* , No Forecast)	3.00 ± 1.80	12.13 ± 5.57	378.9 ± 14.6	2.52 ± 0.47	8 / 20 (40%)	10 / 20 (6.1 s)
Ablation 2	MPPI + Lagged-UA Filter (No Forecast)	2.61 ± 1.34	9.82 ± 5.39	373.4 ± 13.9	2.58 ± 0.35	6 / 20 (30%)	13 / 20 (10.8 s)
Baseline 2b	NMPC (With Forecast Preview)	1.47 ± 0.85	2.82 ± 2.93	356.8 ± 7.6	6.42 ± 1.03	1 / 20 (5%)	19 / 20 (2.7 s)
Ablation 3a	MPPI + Filter (With Forecast Preview)	1.75 ± 1.00	3.10 ± 2.98	357.3 ± 7.7	2.80 ± 0.24	1 / 20 (5%)	18 / 20 (5.6 s)
Ablation 3b	Rule Supervisor + MPPI	1.45 ± 0.68	9.72 ± 4.92	382.6 ± 17.6	3.82 ± 0.33	8 / 20 (40%)	12 / 20 (12.4 s)
Full System	Anticipatory Schedule (Full Architecture)	0.94 ± 0.41	2.15 ± 2.55	355.4 ± 6.1	3.39 ± 0.27	1 / 20 (5%)	19 / 20 (2.4 s)

Key Findings: Linear MPC ($q_T = 10$), NMPC, and MPPI+filter establish an empirical feedback floor of 6/20 trips, failing on the identical scenarios (Scenarios 4, 5, 8, 16, 18, 20). All baseline trips and Scenario 18 occur 4–31 s post-surge during recovery. Scenario 18 requires **6.0 s of advance lead in open loop** (the sole scenario requiring > 5.0 s). Closed-loop runs are evaluated over a standardized 60.0 s window; baseline trips occur as late as $t = 56$ s, and six surviving baseline runs remain unsettled at the 60 s boundary. Trajectory inspection on rule supervisor trip cases (Scenarios 2, 3, 9) shows reactor quenching to $T_{\min} = 339\text{--}344$ K (vs. ≈ 349 K for Full system), allowing reactant to accumulate to $C_A = 0.555\text{--}0.576$ mol/L at surge end and climbing to 0.578–0.597 mol/L (with Scenario 9 reaching a peak of 0.578 mol/L) before exploding into thermal blowout. Crucially, the rule supervisor survives Scenario 18 ($T_{\min} = 348.8$ K) while preview MPPI and Full system trip at 3.5 s lead.

6.2 Deconstructing the Supervisory Advantage

Ablation across the 19 common surviving scenarios isolates the quantitative mechanism behind the supervisory schedule:

Configuration	Non-Trip RMSE (K)	Physical Mechanism
MPPI + Forecast Preview	1.75 ± 1.00	Unconstrained optimal preview tracking
+ Ramp Cap Only	1.12 ± 0.48	Modulates pre-cooling to 282 K, consistent with avoiding reaction quenching
+ Elevated Barrier Weights Only	1.33 ± 0.72	Stiff penalty keeps state strictly within safe envelope
Both (Anticipatory Supervisory Schedule)	0.94 ± 0.41	Coordinated pre-cooling containment (gain: 0.81 K, $p = 0.013$)

Across the 19 shared survivors, the gain is 0.81 K (paired $t\ p = 0.013$, Wilcoxon $p = 0.0017$, Holm-adjusted $p = 0.063$). Overall difference across 20 scenarios is +0.95 K ($p = 0.0069$). Only the ramp-only comparison survives Holm ($p = 0.0009$). Results are stable across 5 MPPI sampling streams.

6.3 Reproducibility & Computational Budget

All tables were generated using a pure NumPy implementation (`benchmark_cstr_unified.py`):

- Scenario generation with `np.random.seed(42)`.
- Sensor noise: `default_rng(1000 + i)`; MPPI sampling: `default_rng(10000 + 100*rep + seed)`.
- NMPC: L-BFGS-B, `maxiter = 10`, warm-started (not converged at each step).
- Oracle verification: `python benchmark_cstr_unified.py --oracle` directly reproduces the Section 4.1 200 s hold-280 K sweep and per-scenario open-loop lead requirements at 50 Hz.
- Evaluation Window: Standardized closed-loop evaluation window of $t_{\text{total}} = 60.0$ s per scenario; full 20-scenario suite executes in approximately 12 minutes on a dual-core CPU.

7. Industrial Deployment Limits & Conclusion

7.1 Industrial Deployment Limits

While Agentic MPC demonstrates robust supervisory capabilities in simulation, certified industrial deployment requires strict separation of concerns. Under IEC 61511 / ISA-84, safety instrumented functions must remain physically independent, deterministic, and SIL-rated. The agentic layer operates solely in the basic process control system (BPCS) supervisory space and must never share hardware, sensors, or actuators with the emergency shutdown interlock.

Perfect-Forecast Assumption: All preview controllers evaluated in Section 6 receive an exact preview of disturbance onset, duration, and magnitude. In industrial plant deployments, upstream analyzers and operator advisories exhibit lead-time jitter, amplitude estimation errors, and false positives. Characterizing closed-loop robustness under imperfect, noisy, or delayed forecast advisories is an essential direction for future validation.

Evaluation Horizon Limitations: The closed-loop benchmark tests were evaluated over a 60 s time horizon. Because open-loop sweeps reveal delayed thermal re-ignition transients extending up to 83 s when unreacted feed accumulates (Section 4.1), and reactive baselines exhibited trips as late as 56 s with six surviving baseline runs unsettled at 60 s, extending the closed-loop evaluation window to 120–200 s is recommended to capture full asymptotic settling in industrial certification.

7.2 Conclusion

This simulation case study demonstrates that preview lookahead and an anticipatory pre-cooling ramp stabilize an exothermic CSTR under severe kinetic surges. A naive rule-based supervisor that commands flat maximum cooling frequently quenches the reactor, accumulates reactant, and triggers delayed blowout. Modulating a pre-cooling ramp to 282 K is consistent with avoiding reaction quenching while maintaining thermal absorption margin.

We present the Agentic MPC framework as a proposed supervisory architecture for qualitative plant operations. Rather than acting as an evaluated numerical optimizer in these experiments, the LLM layer provides the architectural design pattern for interpreting unstructured operator directives and enforcing formal safety contracts in future human-in-the-loop autonomous plant workflows.

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